

CHE 230: Molecular Systems in Chemical Engineering Homework 3

Due: 4/01/2026

CHE 230 Time: 11:59pm

1. a. Derive the lever rule. (5 points)

Overall composition = C_0

Phase composition = C_α and C_β

mass fractions = w_α and w_β

$$w_\alpha + w_\beta = 1$$

$$C_0 = w_\alpha C_\alpha + w_\beta C_\beta$$

substitute $w_\beta = 1 - w_\alpha$

$$C_0 = w_\alpha C_\alpha + (1 - w_\alpha) C_\beta$$

$$C_0 = w_\alpha C_\alpha + C_\beta - w_\alpha C_\beta$$

$$C_0 - C_\beta = w_\alpha (C_\alpha - C_\beta)$$

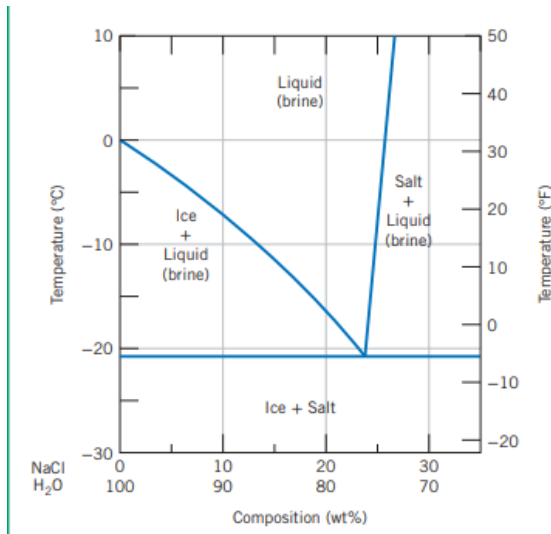
$$w_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha}$$

$$w_\beta = 1 - w_\alpha$$

$$w_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha}$$



b. The following is a portion of the H_2O - NaCl phase diagram: (5 points)



(i) Using this diagram, briefly explain how spreading salt on ice that is at a temperature below 0°C (32°F) can cause the ice to melt.

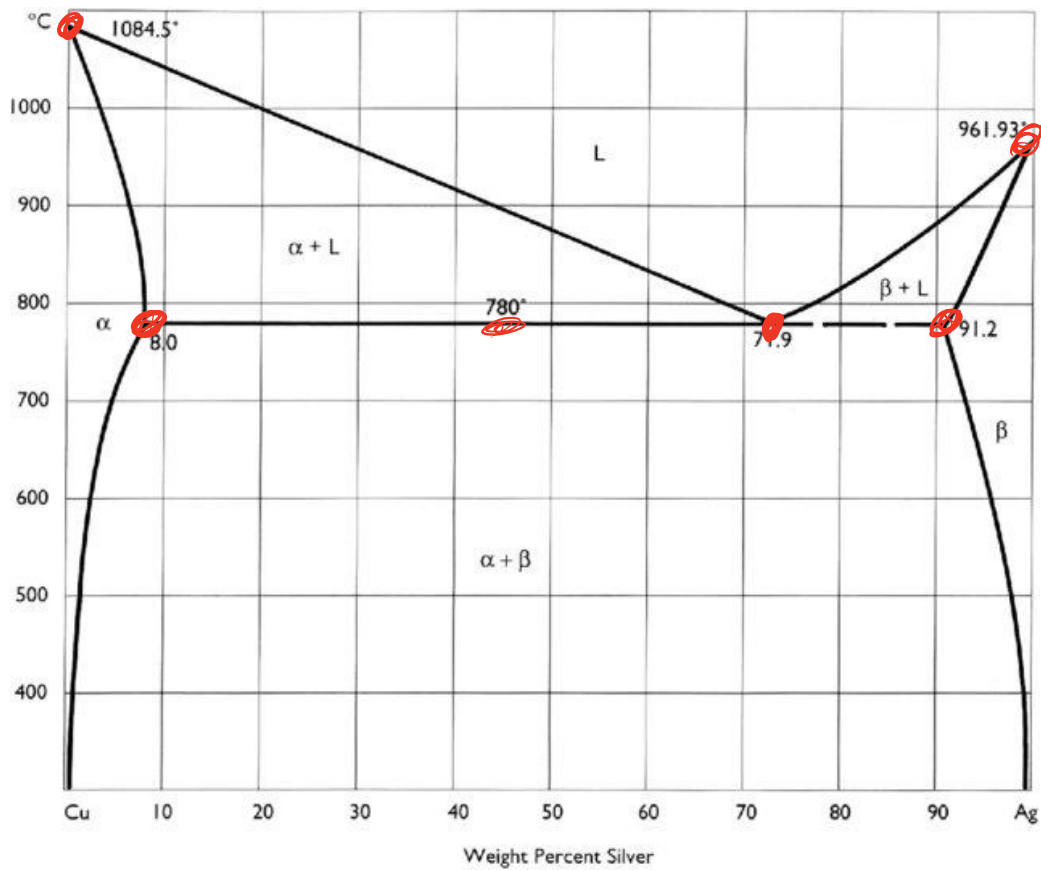
(ii) At what temperature is salt no longer useful in causing ice to melt?

i) Salt dissolves into a small amount of liquid water on the ice surface and forms a brine. That brine has a lower freezing temperature than pure water, so ice can melt even when the temperature is below 0°C.

ii) salt stops being useful once the temperature is below the eutectic temperature which it is about -21°C (approximately)

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2. 90 wt% Ag-10 wt% Cu alloy is heated to a temperature within the β + liquid phase region. If the composition of the liquid phase is 85 wt% Ag, determine: **(14 points)**
- (a) The temperature of the alloy
 - (b) The composition of the β phase
 - (c) The mass fractions of both phases



temperature of the alloy

a) $C_0 = 90 \text{ wt\% Ag}$ Overall alloy composition

$C_L = 85 \text{ wt\% Ag}$ Liquid composition

• $(71.9, 780^\circ\text{C})$

• $(100, 961.93)$

$$T = 780 + \frac{85 - 71.9}{100 - 71.9} (961.93 - 780)$$

$$T = 864.8^\circ\text{C}$$

b) Composition of the β phase

$(71.9, 780^\circ\text{C})$

$(100, 961.93^\circ\text{C})$

$$T = 864.8^\circ\text{C}$$

$$C_\beta = 91.2 + \frac{864.8 - 780}{961.93 - 780} (100 - 91.2)$$

$$C_\beta = 95.3 \text{ wt\% Ag}$$

$$4.7 \text{ wt\% Cu}$$

$$c) W_L = \frac{C_\beta - C_0}{C_\beta - C_L}$$

$$W_\beta = \frac{C_0 - C_L}{C_\beta - C_L}$$

$$W_L = \frac{95.3 - 90}{95.3 - 85} = 0.515$$

$$W_\beta = \frac{90 - 85}{95.3 - 85} = 0.485$$

$$W_L = 51.5\%$$

$$W_\beta = 48.5\%$$

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3. a. Explain the phenomenon of temperature change from a molecular perspective and why it is that some materials have higher heat capacity than others (6 points).

Temperature is the average kinetic energy of particles. Materials with higher heat capacity need more energy to raise their temperature because they can store energy in more internal motions.

- b. Explain why a brass lid ring on a glass canning jar loosens when heated (4 points).

Heating makes the brass lid expand increasing its diameter and loosening it from the glass jar.

- c. A brass rod is to be used in an application requiring its ends to be held rigid. If the rod is stress-free at room temperature (20°C or 68°F), what is the maximum temperature to which the rod may be heated without exceeding a compressive stress of 172 MPa (25,000 psi)?

Assume the following properties for brass:

- Modulus of elasticity (E) = 100 GPa (14.6×10^6 psi)
- Linear coefficient of thermal expansion (α) = $20.0 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ ($11.1 \times 10^{-6} \text{ } ^\circ\text{F}^{-1}$) (8 points)

$$\sigma = E \alpha \Delta T$$
$$E = 100 \text{ GPa} \quad \alpha = 20 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$$
$$\sigma = 172 \text{ MPa}$$
$$T_i = 20^\circ\text{C}$$

$$\Delta T = \frac{\sigma}{\epsilon \alpha}$$

$$\Delta T = \frac{(72 \times 10^6)}{(100 \times 10^9)(20 \times 10^{-6})} = 86^\circ\text{C}$$

$$T_{\max} = 20 + 86 = 106^\circ\text{C}$$

4. An aluminum wire 10 m (32.8 ft) long is cooled from 38°C to -1°C (100°F to 30°F). How much change in length will it experience? (6 points)

$$L_0 = 10\text{m} \quad T_i = 38^\circ\text{C} \quad T_f = -1^\circ\text{C}$$

$$\Delta T = -39^\circ\text{C}$$

$$\Delta L = \alpha L_0 \Delta T$$

$$\alpha = 23.1 \times 10^{-6} \text{C}^{-1}$$

$$\Delta L = (23.1 \times 10^{-6})(10)(-39)$$

$$\Delta L = -0.009009\text{m}$$

$$\Delta L = -9.0 \times 10^{-3}\text{m} \leftarrow \text{Length decreases}$$

5. Compute the electrical conductivity of a cylindrical silicon specimen 5.1 mm (0.2 in.) in diameter and 51 mm (2 in.) in length in which a current of 0.1 A passes in an axial direction. A voltage of 12.5 V is measured across two probes that are separated by 38 mm (1.5 in.).

- (b) Compute the resistance over the entire 51 mm (2 in.) of the specimen (8 points).

a) $d = 5.1\text{mm} = 0.0051\text{m}$

$$L = 51\text{mm} = 0.051\text{m}$$

$$I = 0.1\text{A}$$

$$L = 38\text{mm} = 0.038\text{m}$$

$$V = 12.5\text{V}$$

$$A = \frac{\pi d^2}{4} = \frac{\pi (0.0051)^2}{4}$$

$$J = \frac{I}{A}$$

$$E = \frac{V}{L} = \frac{12.5}{0.038}$$

$$\sigma = \frac{J}{E}$$

$$\sigma = 14.88\text{S/m}$$

$$\sigma = 14.9\text{S/m}$$

$$b) R_{51} = \frac{51}{38} \left(\frac{12.5}{0.1} \right)$$

$$R_{51} = \frac{51}{38} (125)$$

$$R_{51} = 167.8 \Omega$$

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6. An n-type semiconductor is known to have an electron concentration of $3 \times 10^{18} \text{ m}^{-3}$. If the electron drift velocity is 100 m/s in an electric field of 500 V/m, calculate the conductivity of this material (4 points).

$$n = 3 \times 10^{18} \text{ m}^{-3}$$

$$V_d = 100 \text{ m/s}$$

$$E = 500 \text{ V/m}$$

$$\mu_e = \frac{V_d}{E} = \frac{100}{500} = 0.2 \frac{\text{m}^2}{\text{V} \cdot \text{s}}$$

$$\sigma = nq\mu_e$$

$$q = 1.602 \times 10^{-19} \text{ C}$$

$$\sigma = (3 \times 10^{18}) (1.602 \times 10^{-19}) (0.2)$$

$$\sigma = 0.04612 \text{ S/m}$$

$$\sigma = 4.61 \times 10^{-2} \text{ S/m}$$

